

OPENWAM: An Open Framework for Composable World-Action Models

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World-action models (WAMs) couple future prediction with robot control, yet existing systems often vary the video backbone, interaction structure, supervision, and inference procedure simultaneously, making their design choices difficult to compare. We introduce **OPENWAM**, an open world-action modeling framework built around a common causal robot-video foundation and configurable video-action interaction. Starting from Wan2.2-5B, we perform causal robot-video pretraining on over 10,000 hours of video, then integrate an action expert through a shared Mixture-of-Transformers architecture that supports joint, video-then-action, action-then-video, and decoupled generation. **OPENWAM** achieves high success rates on four LIBERO suites and real-world bimanual tasks; robot-video training with causal adaptation improves VTA success on LIBERO-Long from 68.4% to 97.8%. The same configurable architecture naturally extends to inverse and forward dynamics, allowing us to study how counterfactual transitions improve independently trained dynamics models beyond demonstrations alone. When only the video predictor is adapted to a new task, a frozen local-context inverse dynamics model trained on counterfactual data and demonstrations achieves 84.0% mean success across four held-out LIBERO-90 tasks, compared with 47.0% for a full-context inverse model and 21.5% for a local-context model trained only on demonstrations. For forward dynamics, counterfactual supervision reduces RGB prediction error by 34.5% and raises outcome identification from 21.1% to 71.3% among 16 same-state outcomes. **OPENWAM** provides a common testbed for comparing WAM interaction designs and for studying dynamics learning from video data beyond successful demonstrations.

Project Page: <https://openwam.stanford.edu>

Code: <https://github.com/OpenWAM/OpenWAM>

Release: Project page & code: June 4, 2026 Paper: October 6, 2026



1. Introduction

World-action models (WAMs) integrate future-observation prediction with robot control. Recent systems such as DreamZero [90], DVA [71] and LingBot-VA [43, 97] use pretrained video models to couple future-observation prediction with robot action generation. However, differences in their backbones, training data, objectives, and inference procedures make it difficult to isolate the effect of video-action interaction.

Two challenges arise when studying these systems systematically. First, video-action interaction spans multiple choices: video and actions can be generated jointly, sequentially, or through restricted cross-modal connections. A shared architecture and evaluation protocol would allow these interaction patterns and generation orders to be compared under controlled conditions. Second, successful demonstrations cover only a limited subset of possible action-outcome relationships. Other interactions, including

failures, exploratory behavior, and counterfactual rollouts, can provide additional supervision about local dynamics. However, it remains unclear whether such data can improve the reuse of an action-grounding component when the visual predictor is adapted to a new task. We study this question by independently training local-context dynamics models and evaluating their transfer across adapted video predictors.

To this end, we introduce **OPENWAM**, an open framework and model series for controlled world-action modeling, summarized in Figure 1. Starting from a pretrained general-purpose video model, we adapt its visual prior through causal robot-video training without action labels, then introduce an action branch through a Mixture-of-Transformers (MoT) architecture [49]. The resulting model supports joint, sequential, and restricted video-action interaction. These configurations share the video initialization, representations, and evaluation pipeline, allowing us to compare how their interaction patterns affect the control performance.

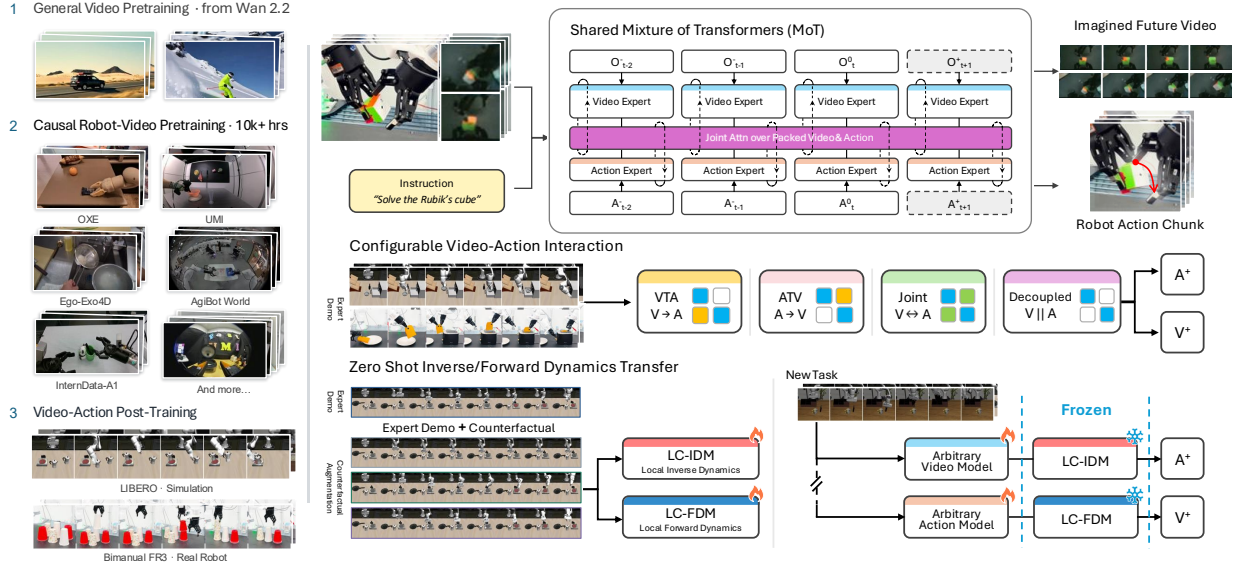


Figure 1. **Overview of OPENWAM.** **Left:** Starting from Wan2.2 [75], causal pretraining on robot and human-interaction video builds a predictive backbone without action labels, followed by video–action training on simulated and real-robot demonstrations. **Top right:** A shared Mixture-of-Transformers couples temporally aligned video and action streams through joint attention while retaining modality-specific experts. **Middle right:** Configurable cross-modal attention and generation order support joint, video-then-action, action-then-video, and decoupled prediction within the same architecture. **Bottom right:** Local-context inverse and forward dynamics models are trained independently using demonstrations and counterfactual transitions. A frozen IDM translates predictions from compatible task-adapted video models into actions, while an FDM predicts the visual consequences of candidate actions.

Furthermore, OPENWAM trains **local-context inverse and forward models** independently. The inverse model predicts actions from the current observation and a future visual trajectory, while the forward model predicts that trajectory from supplied actions. In simulation, we broaden their training data by executing alternative action sequences from the same initial state. This local-transition formulation is also compatible with failed and exploratory interactions, because it does not require a successful task outcome or a success label. Each recorded transition supplies dynamics supervision regardless of whether it completes the task. At inference, a video predictor can propose a visual trajectory and a separately trained inverse model can translate it into executable actions.

We evaluate robot-video pretraining, closed-loop policy success across video–action interaction modes and dynamics-component reuse. In transfer experiments, task-adapted video predictors are paired with frozen inverse models to compare demonstration-only with counterfactual-augmented supervision and local-context with full-context conditioning. We additionally evaluate forward prediction under alternative actions from the same initial state.

This work makes three contributions:

- **A controlled WAM framework.** OPENWAM combines a causal video backbone with configurable video–action models under shared training and evaluation protocols.
- **Local-context dynamics interfaces.** We independently train inverse- and forward-dynamics models from demonstrations and broader local transitions and evaluate their composition with separately trained predictors.
- **Controlled empirical analysis.** We measure the effects of robot-video pretraining, interaction design, transition coverage, and conditioning context on policy performance and component reuse.

2. Related Work

2.1. Visuomotor Policies and Predictive Control

Behavior cloning provides the standard supervised formulation for visuomotor policy learning, mapping observations and task context directly to demonstrated actions. Action chunking and generative action prediction have become effective control primitives [13, 101], while large-scale vision–language–action models extend this paradigm through language conditioning, pretraining, and cross-task or cross-embodiment learning [6–8, 33, 36, 58, 70].

Predictive control introduces an additional interface by

explicitly modeling future observations or latent states. Early visual-foresight methods combined action-conditioned video prediction with model-predictive control for robotic manipulation [16, 17]. Subsequent work learns latent or visual dynamics to support policy learning, planning, and imagined rollouts [11, 24, 25, 80, 82, 107]. Predictive rollouts have also been used to evaluate policy behavior [22, 47].

Action-conditioned world models provide a complementary perspective by predicting how an environment evolves under supplied or candidate actions. Recent methods use such predictions for simulation, candidate evaluation, proposal refinement, and self-verifying planning [31, 61, 95, 106]. Together, these lines show that predictive models can support action generation, simulation, planning, and evaluation. However, action-conditioned prediction and video-action policy generation are often developed as separate components, leaving open how explicit prediction-action interfaces affect component reuse across tasks.

2.2. Video Foundations and World-Action Architectures

Recent world-action models connect predictive video representations with robot control through several forms of video-action coupling. Joint generation models predict future visual content and actions together [1, 21, 66, 90], while causal video-first formulations generate or represent visual futures before action prediction [43, 71]. Other architectures use specialized video and action pathways with explicit cross-modal exchange [87, 96], or leverage egocentric and large-scale video priors for robot grounding and generalization [41, 62, 85, 105].

A closely related line studies the interface between predictive visual information and executable actions. Predicted visual states or representations can condition inverse dynamics [26, 46, 72], while other approaches translate generated futures through dense correspondence, pseudo-action labeling, intermediate denoising features, or latent representation alignment [34, 39, 55, 74]. These methods differ in where and how predictive visual information enters the action-generation pathway.

Unified formulations instead expose multiple predictive and control modes within a shared system. UWM and UVA [44, 108] support combinations of policy, inverse-dynamics, forward-dynamics, and video prediction objectives, while related systems combine action generation with world prediction, simulation, planning, or evaluation [4, 10, 20, 38, 68].

Recent work further isolates particular choices in the WAM design space. Predictive computation can be reduced, sparsified, reordered, or scheduled asynchronously [28, 29, 48, 59, 69, 77, 89, 94]. Alternative work studies

the representation through which world prediction interacts with control [35, 53, 63, 67, 76, 92, 99], or augments world-action learning with structural, geometric, and tactile supervision [18, 56, 104]. Temporal abstraction, data and model scaling, humanoid control, and long-horizon memory provide additional design axes [45, 52, 100, 103]; recent surveys summarize the rapidly expanding landscape [54, 78].

Concurrent work, also named OpenWAM [79], develops a modular framework for systematic WAM pretraining and design studies. Our focus is complementary: OPENWAM fixes a common causal robot-video foundation and training interface while explicitly varying generation order and future cross-modal visibility, enabling joint, video-then-action, action-then-video, and decoupled interaction to be compared under a shared evaluation protocol. We further study the reuse of independently trained inverse- and forward-dynamics components through explicit trajectory interfaces.

2.3. Dynamics Learning Beyond Successful Demonstrations

Successful demonstrations provide valid transitions, but they cover only the action-outcome relationships selected by the demonstrating policy. A complementary analysis shows that observational demonstrations alone may not identify the consequences of unsupported actions [93]. Classical imitation-learning methods address the resulting distribution shift by collecting supervision on learner-visited states, perturbing expert trajectories, or soliciting corrective interventions [27, 32, 40, 57, 65]. These approaches broaden state coverage beyond clean expert rollouts and improve robustness to execution errors.

More recent work expands supervision through synthetic failures, counterfactual augmentation, autonomous experience, and other non-successful interactions [2, 4, 15, 42, 60, 88, 91, 98]. Such data have been used for recovery, policy improvement, failure understanding, and predictive modeling. Particularly related to our setting, Fail2Progress [30] targets data collection toward observed failures to improve skill-effect models, while UniPi [14] decouples video planning from inverse dynamics and allows the inverse model to learn from a separate, potentially suboptimal dataset.

OPENWAM focuses on a different question: how transition coverage and conditioning scope affect the reuse of independently trained dynamics components after visual task adaptation. We branch alternative action continuations from task-relevant simulator states and use the resulting state-action-outcome transitions as local dynamics supervision.

3. Methodology

We develop OPENWAM as an open framework for composing world prediction and action generation. It combines

three ingredients: a causal video backbone pretrained on robot video without action labels, a common video–action architecture whose cross-modal interaction can be reconfigured, and separately trained inverse and forward dynamics components operating through local trajectory interfaces.

The design supports two forms of composition. At the *architectural* level, we vary how future observation and action streams interact while retaining a common backbone and objective family. At the *component* level, we connect independently trained predictors through explicit future–observation and action–trajectory interfaces. The latter motivates local-context dynamics models that do not require the language and nonlocal history used by a task-conditioned generator.

3.1. World–Action Modeling Formulation

Let $o_t \in \mathcal{O}$, $a_t \in \mathcal{A}$, $q_t \in \mathcal{Q}$, and $\ell \in \mathcal{L}$ denote the visual observation, action, proprioceptive state, and language instruction at time t . For fixed observation- and action-history lengths $K_o \in \mathbb{N}$ and $K_a \in \mathbb{N}$, respectively, we define

$$\begin{aligned} O_t^- &= (o_{t-K_o}, \dots, o_{t-1}), \\ A_t^- &= (a_{t-K_a}, \dots, a_{t-1}), \\ O_t^0 &= o_t. \end{aligned} \quad (1)$$

Let $H \in \mathbb{N}_{>0}$ denote the prediction horizon. The future action and observation trajectories are defined as

$$A_t^+ = (a_t, \dots, a_{t+H-1}), \quad O_t^+ = (o_{t+1}, \dots, o_{t+H}). \quad (2)$$

We collect the information available before predicting the future into the task context

$$C_t = (O_t^-, O_t^0, A_t^-, q_t, \ell).$$

Recorded future trajectories are denoted without hats, whereas generated trajectories are denoted by $\hat{O}_t^+$ and $\hat{A}_t^+$. We omit the time index t when it is unambiguous.

OPENWAM models future observations and actions jointly. For a reference joint distribution, the two chain-rule factorizations are

$$\begin{aligned} p_{\text{WAM}}(O_t^+, A_t^+ | C_t) &= \underbrace{p(O_t^+ | C_t)}_{\text{CVP}} \underbrace{p(A_t^+ | O_t^+, C_t)}_{\text{IDM}} \\ &= \underbrace{p(A_t^+ | C_t)}_{\text{VLA}} \underbrace{p(O_t^+ | A_t^+, C_t)}_{\text{FDM}}. \end{aligned} \quad (3)$$

Causal video prediction (CVP) proposes future observations, a direct vision–language–action (VLA) policy proposes actions, inverse dynamics (IDM) grounds a supplied visual transition into actions, and forward dynamics (FDM) predicts the visual consequence of supplied actions.

These roles give three primary generation routes. *Joint generation* produces future observations and actions concurrently. *Video-then-action (VTA)* first generates a visual future and then conditions action generation on that trajectory. *Action-then-video (ATV)* first proposes actions and then predicts their visual consequences. These factorizations specify modeling interfaces; they do not require all factors to share one trained parameterization or final checkpoint.

3.2. An Open Causal Robot-Video Backbone

We initialize the video backbone from Wan2.2-5B [75] and continue pretraining it on heterogeneous robot-manipulation and interaction videos without action supervision. This stage adapts a general-purpose video prior to robot-centric motion and interaction while preserving a standalone video-prediction interface. The resulting checkpoint is shared across all downstream OPENWAM variants, allowing different video–action interaction strategies to be compared under a common visual initialization. Dataset composition and optimization details are summarized in Sec. 4.1.

Because the original video model is not temporally causal, we introduce chunk-causal prediction during robot-video pretraining. A future chunk may attend to the current and preceding observations and previously generated chunks, but not to later chunks; tokens within a chunk are processed jointly. This provides the temporal direction needed for sequential prediction while retaining within-chunk denoising.

Temporal causality and cross-modal interaction are separate design dimensions. Chunk causality determines which times are visible, while the interaction programs in Sec. 3.3 determine which observation and action streams communicate. The pretrained causal backbone can therefore be used directly as a video predictor or coupled to an action branch without changing its basic predictive interface.

3.3. Configurable Video–Action Interaction Programs

Shared architecture. We couple the pretrained video backbone to a separate action transformer using a Mixture-of-Transformers (MoT) architecture [49]. Video and action trajectories are encoded into temporally aligned streams and processed by modality-specific transformer branches. Interleaved cross-modal attention determines how information flows between them. The backbone, tokenization, and objective family are shared across configurations; future cross-modal visibility and generation order are the principal variables. Figure 2 summarizes the architecture, and Figure 3 the attention masks of four interaction programs together with the two local-context dynamics programs.

Interaction programs. We study four video–action interaction programs under a shared Mixture-of-Transformers

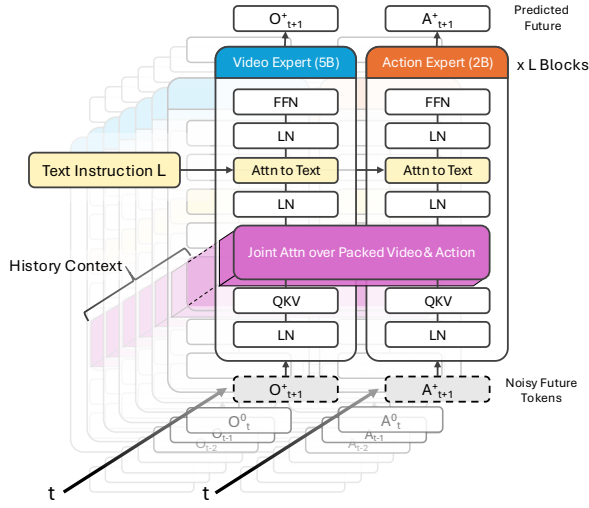


Figure 2. Shared MoT architecture. Two modality-specific branches keep their own normalization, projection and feed-forward parameters and meet in a single attention over the packed video and action tokens; a separate cross-attention consumes the task text. Noisy future tokens enter at the bottom and the preceding chunks sit behind as history context.

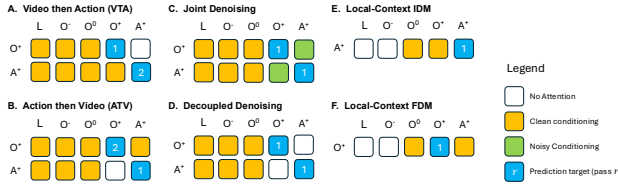


Figure 3. Attention masks the shared architecture exposes: four interaction programs (A–D) and the two local-context dynamics programs (E–F). Columns are the streams a row may condition on, cell colour gives what it attends to there, and the numeral marks the generation pass. E and F carry one row each because each predicts a single stream: the inverse model produces actions, the forward model observations.

backbone. *Joint* denoises future video and action streams concurrently, with bidirectional attention between their noisy future tokens. *VTA* first predicts future video without access to future actions, then predicts the action conditioned on the completed visual trajectory. *ATV* reverses this order: it first predicts the action, then conditions video prediction on the completed action trajectory. *Decoupled* predicts both modalities without future cross-modal attention. All four programs use the same backbone, tokenization, and training objective; they differ only in generation order and future cross-modal attention.

Table 1. LIBERO-90 targets used for component-transfer evaluation.

Task	Target behavior	Transfer challenge
64	Stack bowls; place stack in tray	Novel skill composition
74	Book into caddy’s left compartment	Retarget familiar manipulation
21	Turn on stove; place frying pan on it	Object and grasp change
45	Same objective as Task 21	Additional scene and layout shift

Shared flow-matching objective. All programs use the same latent flow-matching [50] construction. For target modality $P \in \{O, A\}$, let Z_P^* be its clean target latent. With $\tau \sim \mathcal{U}[0, 1]$ and $\epsilon_P \sim \mathcal{N}(0, I)$,

$$Z_{P,\tau} = (1 - \tau)\epsilon_P + \tau Z_P^*, \quad u_P = Z_P^* - \epsilon_P. \quad (4)$$

An active prediction stage minimizes

$$\mathcal{L}_P = \mathbb{E} \left[\|v_{\theta,P}(Z_{P,\tau} | c_P, \tau) - u_P\|_2^2 \right], \quad (5)$$

where c_P denotes the history, side information, and cross-modal trajectory made visible by the interaction program. Programs with multiple targets sum the corresponding losses. Observation and action trajectories used as conditioning are processed through their respective modality encoders.

During training, a sequential program uses teacher forcing: the second generation pass conditions on the recorded cross-modal trajectory rather than on an output sampled by the first pass. At inference, it conditions on the completed trajectory generated by the first pass. The same interaction program and temporal alignment are used during training and inference; the source of the conditioning trajectory changes from recorded to generated.

3.4. Counterfactual Learning of Local-Context Dynamics Interfaces

Local-context dynamics interfaces. Architectural composition controls communication within a model, but reusing a separately trained component requires an explicit conditioning boundary. A task-conditioned generator may use language and nonlocal history to specify behavior; a reusable local dynamics component instead relates a supplied transition to its actions, or supplied actions to their consequences.

We therefore define

$$\begin{aligned} p_{\text{IDM}}^{\text{local}} &:= p(A_t^+ | O_t^+, O_t^0, q_t), \\ p_{\text{FDM}}^{\text{local}} &:= p(O_t^+ | A_t^+, O_t^0, q_t). \end{aligned} \quad (6)$$

Local-context IDM grounds a supplied visual trajectory into an action sequence, while local-context FDM predicts the visual consequence of a supplied action sequence. Neither receives language, past actions, or observations preceding t .

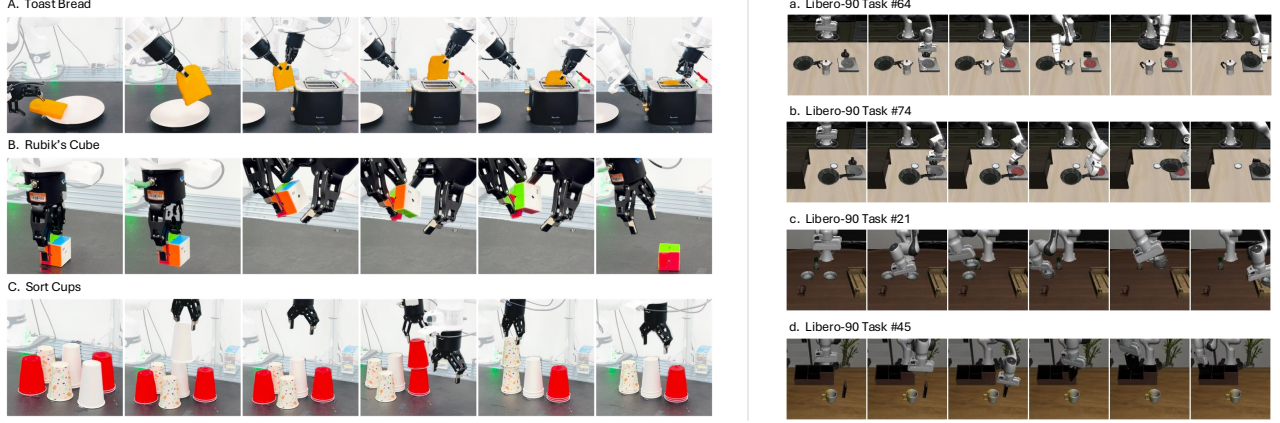


Figure 4. Evaluation tasks. **Left:** the three real-world bimanual tasks of Table 2, each shown as frames sampled along one successful rollout. **Right:** the four LIBERO-90 targets of Table 1 used for component transfer, which lie outside the LIBERO-Long source-task set.

Importantly, conditioning scope and transition supervision are separate design choices. The local-context restriction specifies which variables the dynamics component may consume; it does not determine which transitions are used for learning. Conversely, broader transition supervision can also be applied to a full-context dynamics model. Sec. 4.3 therefore compares full-context and local-context IDMs under matched demonstration-only and mixed supervision.

Local-context conditioning restricts the variables available to the dynamics component; it does not assume universal local sufficiency. Its adequacy depends on whether the information needed for local action–outcome inference is observable from the current observation, proprioceptive state, and supplied transition. We evaluate this interface through local prediction and composed control.

Broader local-transition supervision. A task trajectory induces a local transition $z = (O_t^0, q_t, A_t^+, O_t^+)$. Let $\mathcal{D}_{\text{task}}^{\text{loc}}$ denote the distribution of such transitions obtained from demonstrations. Demonstrations provide valid transitions but contain only the action continuations selected by the policy.

To broaden local supervision, we restore selected simulator states and execute alternative future action sequences. For branch b , this produces $z^{(b)} = (O_t^0, q_t, A_t^{+, (b)}, O_t^{+, (b)})$, where the initial condition is shared and each future observation trajectory results from executing its corresponding action sequence. These alternatives include interactions outside the demonstrated continuation and need not complete the task. The simulator state is used only to control data collection and is not provided to the model.

The paired collection protocol controls the starting state, while the learning objective operates on individual transition

records rather than using an explicit paired-branch loss. Let $\mathcal{D}_{\text{cf}}$ denote the resulting counterfactual transitions. Dynamics models are trained from

$$\mathcal{D}_{\text{dyn}} = (1 - \eta)\mathcal{D}_{\text{task}}^{\text{loc}} + \eta\mathcal{D}_{\text{cf}}, \quad \eta \in [0, 1]. \quad (7)$$

This supports demonstration-only, counterfactual-only, and mixed supervision without changing the conditioning interface. The experimental mixture and collection details are given in Sec. 4.1. The same interface can be applied to failed or exploratory physical-robot transitions when such data are collected.

Local dynamics learning. Using the flow construction in Eq. 4, local-context IDM denoises the action target while conditioning on the observed future, whereas local-context FDM denoises the future observation target while conditioning on the executed action sequence:

$$\begin{aligned} \mathcal{L}_{\text{LC-IDM}} &= \mathbb{E} \left[\|v_{\theta_{\text{ID}}} (Z_{A, \tau} \mid O^0, q_t, O^+, \tau) - u_A\|_2^2 \right], \\ \mathcal{L}_{\text{LC-FDM}} &= \mathbb{E} \left[\|v_{\theta_{\text{FD}}} (Z_{O, \tau} \mid O^0, q_t, A^+, \tau) - u_O\|_2^2 \right]. \end{aligned} \quad (8)$$

Ground-truth future observations condition IDM training but are not video denoising targets; analogously, executed actions condition FDM training.

Inference and component composition. For video-to-action composition, the video predictor passes future variational autoencoder (VAE) video latents to the receiving IDM; transformer hidden states and caches are not transferred across components. The receiving IDM processes the transferred latents through the same modality-conditioning pathway used for recorded conditioning.

At inference, a task-conditioned visual predictor can be paired with an independently trained local-context IDM:

$$\hat{O}_t^+ \sim p_\phi^{\text{CVP}}(\cdot | \mathcal{C}_t), \quad \hat{A}_t^+ \sim p_{\theta_{\text{ID}}}^{\text{LC-IDM}}(\cdot | \hat{O}_t^+, O_t^0, q_t). \quad (9)$$

Task context determines the desired future through the visual predictor, while the receiving IDM receives only the generated future trajectory and the current local inputs.

The complementary composition pairs an action generator with local-context FDM:

$$\hat{A}_t^+ \sim p_\psi^{\text{VLA}}(\cdot | \mathcal{C}_t), \quad \hat{O}_t^+ \sim p_{\theta_{\text{FD}}}^{\text{LC-FDM}}(\cdot | \hat{A}_t^+, O_t^0, q_t). \quad (10)$$

The action sequence may equivalently be supplied by an external policy. These are learned compositions: separately trained components need not recover factors of one identical learned joint distribution. Their common trajectory interfaces instead make the components explicitly replaceable, and Sec. 4.3 evaluates when such replacement succeeds.

4. Experiments

We evaluate OPENWAM along three dimensions: *native policy performance*, *component composability*, and *local dynamics modeling*. We test whether the shared robot-video foundation supports different video–action interaction programs, whether independently trained action components remain useful when paired with adapted visual predictors, and whether local-context forward dynamics models action-dependent visual outcomes.

4.1. Experimental Setup

Robot-video pretraining and simulation. We initialize the visual backbone from Wan2.2-5B [75] and further pre-train it without action supervision on heterogeneous robot and human-interaction video, including UMI/MV-UMI [12, 64], AgiBot World [9], RoboMIND [81], InternData-A1 [73], RoboCOIN [84], FastUMI [102], Ego-Exo4D [19], and Open X-Embodiment [58]. Unless otherwise stated, downstream OPENWAM variants use this common initialization.

The four interaction programs share the same dual-stream MoT backbone and downstream training recipe; they differ in generation order and future cross-modal attention. Video is processed in four-frame latent chunks, each predicting a 16-step action chunk, with proprioception provided as a per-chunk conditioning signal. Actions consist of end-effector position, axis-angle rotation, and a gripper command. We use a two-stage training procedure: causal robot-video pre-training followed by joint video–action training on the target suite.

We evaluate Joint, VTA, ATV, and Decoupled on LIBERO-Object, LIBERO-Goal, LIBERO-Spatial, and

Table 2. Real-world bimanual manipulation tasks.

Task	Objective	Main challenge
Toast Bread	Insert bread and activate toaster	Deformability and precise insertion
Rubik’s Cube	Restore final layer of 2×2 cube	Contact-rich push and rotation
Sort Cups	Sort cups according to color	Variable ordering and sequencing

LIBERO-Long [51]. Component transfer uses LIBERO-Long as the downstream source distribution and four LIBERO-90 targets (Table 1) chosen to represent qualitatively different forms of transfer. These targets are outside the LIBERO-Long source-task set; robot-video pretraining is treated as a separate stage.

Real-world tasks. We evaluate on a bimanual platform comprising two Franka Research 3 arms with parallel-jaw grippers and three RGB views: two wrist-mounted cameras and one centered third-person camera. The three task definitions are listed in Table 2. Figure 4 shows the real-world tasks and the LIBERO-90 transfer targets.

Dynamics supervision. Native task models are trained from task demonstrations. For local dynamics, we additionally construct **LIBERO-Long-CF** by executing alternative action continuations from states encountered in LIBERO-Long demonstrations. Branches sharing an initial state remain in the same train, validation, or test partition. IDM and FDM experiments use demonstration-only, counterfactual-only (CF-only), or mixed supervision. Mixed supervision samples 60% of transitions from $\mathcal{D}_{\text{cf}}$ and 40% from $\mathcal{D}_{\text{task}}^{\text{loc}}$, corresponding to $\eta = 0.6$ in Eq. 7.

Evaluation. Closed-loop task success is the primary policy and composition metric. For forward dynamics, we report visual prediction error and outcome identification accuracy. Given K realized futures from the same state, the metric tests whether the prediction is closest in RGB MSE to its matching outcome. We report both pairwise identification ($K = 2$) and identification among all 16 action branches ($K = 16$).

4.2. Native Policy Performance

Broad LIBERO evaluation. We compare the four interaction programs across the LIBERO suites and report the average success rate over three random seeds in Table 3. The principal OPENWAM configurations achieve strong performance across all four suites, establishing that the shared foundation supports multiple world–action generation orders.

Backbone initialization ablation. Holding the VTA architecture, task supervision, and downstream training procedure fixed, the original Wan2.2-5B initialization reaches 68.4%

Table 3. Closed-loop success (%) on four LIBERO suites. Baselines are previously reported results.

Method	Object	Goal	Spatial	Long	Mean
OpenVLA [36]	88.4	79.2	84.7	53.7	76.5
OpenVLA-OFT [37]	98.4	97.9	97.6	94.5	97.1
π_0 [6]	98.8	95.8	96.8	85.2	94.1
$\pi_{0.5}$ [33]	98.2	98.0	98.8	92.4	96.9
GR00T-N1 [5]	97.6	93.0	94.4	90.6	93.9
Motus [3]	99.8	96.6	96.8	97.6	97.7
Fast-WAM [94]	100.0	97.0	98.2	95.2	97.6
LingBot-VA [43]	99.6	97.2	98.5	98.5	98.5
OPENWAM-VTA	99.4 $\pm$ 0.3	98.4 $\pm$ 0.5	98.6 $\pm$ 0.2	97.8 $\pm$ 0.4	98.6
OPENWAM-ATV	98.0 $\pm$ 0.4	97.2 $\pm$ 0.2	96.6 $\pm$ 0.6	95.4 $\pm$ 0.3	96.8
OPENWAM-Joint	98.2 $\pm$ 0.2	97.8 $\pm$ 0.4	97.6 $\pm$ 0.3	96.6 $\pm$ 0.5	97.6
OPENWAM-Decoupled	99.0 $\pm$ 0.3	98.0 $\pm$ 0.2	97.8 $\pm$ 0.5	97.0 $\pm$ 0.4	98.0

Table 4. Closed-loop success (%) on real-world bimanual manipulation tasks.

Method	Toast	Cube	Cups	Mean
OPENWAM-VTA	92.0	90.0	94.4	92.1
OPENWAM-Joint	90.0	94.0	91.7	91.9

success on LIBERO-Long, while our robot-video-pretrained causal backbone reaches 97.8% (+29.4 points). This comparison measures the combined effect of robot-video pretraining and causal adaptation, rather than attributing the gain to either factor individually.

Real-world native policies. Table 4 reports closed-loop success for VTA and Joint on the three bimanual tasks of Table 2, which test the same framework under multi-view, deformable, contact-rich, and multi-object manipulation.

4.3. Component Composability

We next study whether action-grounding components remain useful when the visual predictor changes. Full-context and local-context IDMs are evaluated under transition-supervision regimes, separating the effects of broader local-transition data from the restriction of task-level context.

Frozen action-component reuse. For each LIBERO-90 target, the visual producer is taken from a task-tuned VTA model while source-trained action components remain frozen. Thus, the experiment measures reuse of an action component after visual-task adaptation, rather than target-task learning without action supervision. All reused executors are evaluated with the same target predictor and rollout protocol. Table 6 reports source-task composition on LIBERO-Long.

Conditioning scope and supervision interact strongly (Tables 6 and 5). Mixed supervision raises mean target success from 21.5% to 84.0% for local-context IDMs, compared with 44.5% to 47.0% for full-context IDMs. Under matched mixed supervision, local conditioning improves tar-

get success by 37.0 percentage points while retaining 94.4% source-task success. Counterfactual-only and mixed supervision both substantially improve local-context IDM transfer, achieving comparable mean success of 84.5% and 84.0%, respectively, across the four LIBERO-90 targets. These results show that broader transition coverage supports the reuse of local-context dynamics components after visual task adaptation.

The per-task results also reveal uneven transfer. The demonstration-only full-context IDM performs well on Tasks 64 and 74 but obtains 0% success on Tasks 21 and 45, whereas CF-only and mixed local-context IDMs remain effective across all four targets. This contrast highlights their more consistent reuse across the evaluated task changes.

4.4. Action-Conditioned Forward Dynamics

We evaluate local-context FDMs on 2,560 counterfactual futures from 160 contexts across ten LIBERO tasks, with 16 action branches per context. Models share the same initial observations, actions, and realized futures. In addition to RGB prediction error, we measure whether the predicted future preserves the identity of its conditioning action through outcome identification at $K = 2$ and $K = 16$.

As shown in Table 7, CF-only and mixed supervision reduce RGB MSE by 34.5% and 33.0%, respectively, relative to demonstration-only training. CF-only supervision raises outcome accuracy from 68.3% to 93.6% for $K = 2$, and from 21.1% to 71.3% for $K = 16$. Mixed supervision shows a similar improvement, reaching 91.8% and 67.7%, respectively. These results show that broader transition supervision improves both visual prediction accuracy and the ability to distinguish action-dependent futures.

5. Discussion and Conclusion

We presented OPENWAM, an open framework and pre-trained foundation spanning video prediction, robot control, and local dynamics modeling. Its shared causal backbone supports joint, sequential, and decoupled video-action generation within a common training and evaluation stack. Strong performance across four LIBERO suites and real-world bimanual tasks demonstrates the framework’s control capabilities, while the initialization ablation establishes the substantial benefit of robot-video pretraining. OPENWAM thus supports multiple effective modeling configurations rather than a single specialized policy.

Beyond native control, the framework supports independently trained inverse and forward dynamics components. In the transfer study, counterfactual-only and mixed supervision both substantially improve the reuse of local-context IDMs after visual task adaptation, while retaining strong source-task performance. Counterfactual FDM supervision

Table 5. Frozen inverse dynamics transfer to four LIBERO-90 targets. Target rows use task-tuned video predictors while source-trained IDMs remain frozen. Mean is averaged over the four target tasks.

Video / policy	Action component (frozen)	64 Composition	74 Retarget	21 Object/grasp	45 Scene shift	Transfer mean
Task-tuned VTA (finetuned policy reference)		94%	86%	92%	100%	93.0%
Task-tuned video	Full-context IDM, demo-only	88%	90%	0%	0%	44.5%
Task-tuned video	Full-context IDM, mixed	54%	92%	4%	38%	47.0%
Task-tuned video	Local-context IDM, demo-only	36%	50%	0%	0%	21.5%
Task-tuned video	Local-context IDM, CF-only	80%	92%	90%	76%	84.5%
Task-tuned video	Local-context IDM, mixed	74%	86%	88%	88%	84.0%
Original VTA (unadapted policy reference)		42%	0%	0%	0%	10.5%

Table 6. Source-task composition on LIBERO-Long, all action components evaluated with LIBERO-Long video predictor.

Action component	Supervision	Success (%)
Full-context IDM	Demo-only (Native VTA)	97.8
Full-context IDM	Mixed	92.2
Local-context IDM	Demo-only	25.8
Local-context IDM	CF-only	90.6
Local-context IDM	Mixed	94.4

Table 7. Local-context FDM prediction on counterfactual transitions. MSE values are reported in units of 10^{-3} . Outcome accuracy measures identification of the matching realized future by RGB MSE among K same-state outcomes.

Supervision	RGB MSE $\downarrow$	Acc. $K=2 \uparrow$	Acc. $K=16 \uparrow$
Demo-only	14.35	68.3	21.1
CF-only	9.40	93.6	71.3
Mixed	9.62	91.8	67.7

also improves both visual prediction accuracy and identification of action-dependent outcomes.

Several questions remain open. Composability is established primarily in simulation; demonstrating comparable reuse on real robots will require more data-efficient physical dynamics supervision. Our target-task experiments adapt the video predictor and therefore do not establish zero-shot transfer of the complete video–action model. Appendix E shows that policy, IDM, and FDM objectives can be trained jointly in one checkpoint, but we have not yet shown that a single multi-objective model can match the strongest specialized OPENWAM variants across all modes. The present FDM study also focuses on local, short-horizon prediction, leaving long-horizon planning and physical deployment as natural extensions.

OPENWAM brings these capabilities into one open foundation: strong policies, configurable prediction–action interaction, and independently reusable dynamics. It provides both capable world–action models and a common basis for

building, comparing, and composing the next generation.

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Appendix

A. Implementation Details

This section provides implementation and evaluation details for the four video–action interaction programs studied in the main paper, including model architecture, downstream training, generation, and closed-loop evaluation.

A.1. Model Architecture and Representation

Dual-stream backbone. We use a 30-layer dual-stream Mixture-of-Transformers (MoT) architecture with modality-specific video and action experts. The video expert has a hidden dimension of 3072, 24 attention heads with a head dimension of 128, and an FFN dimension of 14336. The action expert has a hidden dimension of 2048 and an FFN dimension of 8192, while retaining the same attention configuration of 24 heads with a head dimension of 128. Consequently, its query, key, and value projections each map the 2048-dimensional hidden state to 3072 dimensions, and the attention output is projected back to 2048 dimensions. Cross-modal information exchange is controlled by the future cross-modal attention pattern associated with each interaction program. Task text conditions both streams through a separate cross-attention layer.

Visual observations. Each observation contains an agent-view image and a wrist-camera image, both at 128×128 resolution. The two views are encoded independently into $8 \times 8 \times 48$ VAE latents and concatenated along the spatial width dimension to form an $8 \times 16 \times 48$ latent canvas. Using a $1 \times 2 \times 2$ temporal-spatial patch size, each latent frame contains 32 video tokens. The video VAE temporally compresses every four environment steps into one latent video frame. Each generation chunk spans four future latent frames, corresponding to 16 environment steps.

Actions and proprioception. The policy predicts a 16-step action chunk aligned with the same temporal horizon, with four action steps corresponding to each latent video frame. Each action has seven dimensions: three for end-effector position, three for axis-angle rotation, and one for the gripper command. Each chunk is conditioned on an eight-dimensional proprioceptive state comprising the 3D end-effector position, a 3D axis-angle orientation converted from the simulator quaternion, and two gripper joint positions. The proprioceptive state is projected to the corresponding hidden dimension of each stream and added to its hidden representation rather than appended as a sequence token.

A.2. Downstream Training

The four interaction programs start from the same robot-video-pretrained causal backbone. The action expert is ini-

tialized from width-adapted copies of the corresponding pre-trained video-expert layers, while action-specific input/output and conditioning layers are initialized separately. The video and action experts are then jointly optimized during downstream training. All programs use the same data, optimization settings, and flow-matching objective; they differ only in generation order and future cross-modal attention.

For the downstream runs examined here, training uses AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1, a learning rate of 10^{-5} , with 10 warmup steps followed by a constant learning-rate schedule. We use a per-device batch size of 1 and accumulate gradients over 10 steps, yielding an effective batch size of 10. Gradient norms are clipped at 2.0, and training uses bf16 mixed precision. We train all downstream models for 10,000 optimization steps.

Flow-matching objective. Both streams are trained with the rectified-flow objective in Eq. (4). For implementation, we parameterize the noise level as $\sigma = 1 - \tau$, such that $\sigma = 0$ denotes clean data and $\sigma = 1$ denotes pure noise. We uniformly sample from a 1,000-level base noise grid and apply the modality-specific shift

$$S_s(\sigma) = \frac{s\sigma}{1 + (s-1)\sigma}, \quad s_V = 5.0, \quad s_A = 1.0, \quad (11)$$

for video and action, respectively.

We additionally apply timestep-dependent loss weighting,

$$w_P(\sigma) \propto \exp\left[-2\left(\sigma - \frac{1}{2}\right)^2\right] - e^{-1/2}, \quad (12)$$

where the weights are normalized to have unit mean over the corresponding timestep grid. This emphasizes intermediate noise levels while assigning zero weight to the pure-noise endpoint.

The flow-matching loss is applied only to predicted future video latents and valid action entries. Video losses are normalized over the supervised latent elements, while action losses are averaged over the action horizon after masking invalid action dimensions. The two modality losses are combined with equal weight,

$$\mathcal{L} = \mathcal{L}_V + \mathcal{L}_A. \quad (13)$$

A.3. Video–Action Interaction Programs

The four programs differ in how predicted video and action trajectories interact within a generation chunk. Each program also receives the observed history and task instruction.

Joint. Video and action trajectories are denoised concurrently. During generation, each stream can attend to the other stream’s evolving prediction, allowing bidirectional interaction before either trajectory is complete.

Video-to-action (VTA). The model first generates the future video trajectory without conditioning on future actions. It then generates the action trajectory conditioned on the generated video trajectory.

Action-to-video (ATV). The model first generates the future action trajectory without conditioning on future video. It then generates the video trajectory conditioned on the generated action trajectory.

Decoupled. Video and action trajectories are generated concurrently, without cross-modal interaction between their future predictions. Both streams can still use the observed history and task instruction.

During training, the second stage of VTA and ATV conditions on the ground-truth trajectory of the first modality. At inference time, this ground-truth trajectory is replaced by the trajectory generated in the first stage. Joint and Decoupled jointly denoise both modalities in a single generation stage, whereas VTA and ATV use two sequential generation stages.

A.4. Inference and Closed-Loop Evaluation

Sampling. We use 20 denoising steps for each modality, with the same modality-specific timestep schedules as in training. Video generation uses classifier-free guidance with scale 5.0, while no classifier-free guidance is applied to actions. Key-value caching is enabled during inference. The history window spans 15 generation chunks, corresponding to 60 latent video frames and the temporally aligned action history.

Action execution. After resetting to the episode’s initial state, the simulator is advanced for five zero-action steps to settle, and the policy is initialized from the final observation. The first generation chunk conditions on this single observation; the same startup convention is used during training, so no additional history frames are required before the first prediction. The policy executes each 16-step action chunk before replanning. Episodes terminate upon task success, after 800 environment steps (including the five initialization steps), or after 50 action chunks, whichever comes first.

Evaluation protocol. For each interaction program, we evaluate three independently trained seeds. Each checkpoint is evaluated on 10 tasks with 50 closed-loop episodes per task, for 500 episodes per seed. Within each task, all interaction programs use the same 50 environment seeds and therefore share the same set of initial conditions. An episode is counted as successful if the simulator’s task-success predicate is satisfied at any environment step, at which point the rollout terminates. We report the mean and standard deviation of the success rate across the three seeds.



Figure 5. Real-world bimanual setup with two Franka FR3 arms, Flexiv Grav grippers, bimanual GELLO teleoperation [83], two wrist-mounted RealSense D405 cameras, and a central Orbbec Femto Mega camera.

A.5. Real Robot Setup

The real-robot setup, as shown in Figure 5, consists of two Franka FR3 arms with Flexiv Grav grippers. We collect demonstrations through bimanual GELLO teleoperation. Two wrist-mounted RealSense D405 cameras record at 1280×720 pixels, and a central Orbbec Femto Mega records at 1920×1080 pixels, all at 30 fps. Joint positions, end-effector poses, gripper widths, and control commands are recorded with timestamps at 30 Hz. In the chunk-relative formulation, each arm’s target poses are expressed relative to its observed pose at the start of the chunk, while gripper widths remain absolute. Each action contains 20 values: three position coordinates, six values representing orientation, and one gripper width for each arm.

For the real-robot experiments, we collect 200 demonstrations each for Toast Bread and Rubik’s Cube, and 180 demonstrations for Sort Cups. Each demonstration is 30 s long. We evaluate the learned policies over 50 closed-loop episodes for Toast Bread and Rubik’s Cube, and 36 episodes for Sort Cups.

B. Robot-Video Pretraining

We adapt the Wan2.2-5B video backbone to robot-centric motion and physical interaction before introducing action prediction. The pretraining corpus combines robot demonstrations, human manipulation video, and synthetic robot trajectories across a broad range of embodiments, viewpoints, objects, and interaction patterns. No action labels are used during this stage.

B.1. Pretraining Corpus

Our robot-video pretraining corpus combines large-scale real-robot manipulation, human interaction video, and synthetic robot trajectories. Table 8 summarizes the source data

Table 8. Data used for causal robot-video pretraining.

Dataset	Trajectories / takes	Hours	Source type
Open X-Embodiment (49 manipulation datasets)	1,300,749	1,911.29	Multi-embodiment robot data
AgiBot World Beta	1,003,672	2,976.4	Real robot manipulation
Ego-Exo4D v2	5,035	221.26	Human interaction
InternData-A1	637,498	7,433.91	Synthetic robot manipulation
RoboCOIN	183,157	1,306.83	Real bimanual manipulation
RoboMIND	107,877	305.5	Multi-embodiment robot data
FastUMI-100K	92,823	461.77	Human-guided manipulation
UMI family	5,430	24.60	Human-guided manipulation

before training-window sampling. The UMI-family entry aggregates UMI, DexUMI [86], UMI on Legs [23], and MV-UMI.

The corpus contains approximately 3.34 million trajectory or take units and 14.64k nominal hours, spanning heterogeneous robot embodiments, collection interfaces, viewpoints, human-object interactions, and synthetic manipulation at scale, providing the visual and interaction prior used by all downstream OPENWAM models.

B.2. Causal Adaptation

Pretraining starts from the Wan2.2-5B checkpoint. We retain the pretrained video representation and optimize the model on the corpus above using the chunk-causal prediction objective described in Sec. 3. Future chunks may attend to the current observation, preceding observations, and previously generated chunks, but not to later chunks. Tokens within each chunk are denoised jointly.

This stage uses video alone. Robot actions, proprioception, and task-specific policy targets are not supplied to the model. The resulting checkpoint serves both as a standalone causal video predictor and as the common visual initialization for the downstream OPENWAM interaction programs.

B.3. Training Compute

Causal robot-video pretraining is performed on 32 NVIDIA B200 GPUs for 14 days. All downstream OPENWAM variants in the main experiments start from the resulting checkpoint unless otherwise specified.

C. LIBERO-Long-CF: Counterfactual Dynamics Data

LIBERO demonstrations contain task-directed behavior and cover only a small fraction of the local state–action–outcome distribution. From each visited state, they provide essentially one successful action continuation. This is well suited to imitation learning but provides limited supervision for inverse and forward dynamics, which must model how alternative actions correspond to alternative physical outcomes.

Table 9. Scale of LIBERO-Long-CF and the source LIBERO-Long demonstration corpus.

Quantity	Demonstrations	LIBERO-Long-CF
Tasks	10	10
Stored sequences	500	32,000
Sequences / task	50	3,200
Controls / sequence	276.2 mean	128
Total controls	138,090	4,096,000
Duration	1.92 h	56.9 h

We construct **LIBERO-Long-CF** by branching from task-relevant states in the ten LIBERO-Long tasks and executing modified future action sequences. The dataset contains 32,000 counterfactual segments, balanced across tasks. Each segment contains 128 future controls and 129 synchronized two-view observations. The original tasks, assets, and simulator physics are preserved.

C.1. Scale and Intervention Coverage

LIBERO-Long-CF contains 4.10 million control records, compared with 138,090 controls in the 500 source demonstrations, increasing local transition supervision by approximately $29.7\times$. At 20 Hz, these correspond to 56.9 and 1.92 control-equivalent hours, respectively.

The branch generator modifies motion magnitude, direction, temporal structure, individual action axes, stochastic arm motion, and gripper behavior. Table 10 summarizes the intervention distribution.

Interventions include stopped and rescaled motion, reversed or redirected translation, translational and rotational pulses, structured control noise, piecewise-random commands, and changes to gripper state and timing. All commands are within LIBERO’s normalized controller range.

C.2. State and Interaction Coverage

Within the dataset, 75% of segments begin from demonstration-prefix states. The remaining 25% begin after an additional executed perturbation, extending the starting-

Table 10. Intervention composition of LIBERO-Long-CF. Categories denote generation recipes and may induce overlapping physical effects.

Intervention family	Fraction (%)
Stop / rescale arm motion	9.4
Reverse / redirect translation	6.3
Axis biases and pulses	12.5
Dedicated yaw perturbation	3.1
Noise / randomized arm controls	12.5
Dedicated gripper interventions	31.3
Random-duration arm / gripper interventions	25.0

Table 11. Physical interaction statistics for LIBERO-Long-CF. Categories overlap.

Measured interaction	Rate (%)
Gripper-object / fixture contact	90.4
Detected grasp	50.0
Object-configuration effect	72.3

state distribution beyond the demonstrated trajectories.

Among characterized perturbed starts, the end-effector state is on average 3.34 cm from the nearest point on the corresponding demonstrated path (median 1.90 cm). Under thresholds of 1 cm translation, 5° rotation, or 5% articulated-joint travel, 61.6% also contain object configurations outside the corresponding demonstrated configurations.

The corpus contains substantial physical interaction. As shown in Table 11, 90.4% of characterized segments contain gripper contact with an object or fixture, 50.0% contain a detected grasp, and 72.3% alter object configuration relative to the same-start reference continuation.

C.3. Representation for Local Dynamics Learning

Each segment contains two synchronized 128×128 RGB streams, 128 seven-dimensional delta-OSC controls, and 129 proprioceptive states. Actions contain three translation channels, three rotation channels, and one gripper command.

Both camera streams are encoded with the OPENWAM video VAE. The encoded sequence contains one boundary observation and 32 future latent frames, with four controls aligned to each future latent frame.

Local-context IDM conditions on the boundary observation, proprioception, and future visual trajectory to predict the action sequence. Local-context FDM conditions on the boundary observation, proprioception, and action sequence to predict the future visual trajectory. Both objectives exclude task language and pre-start history.

Table 12. Effect of video-backbone initialization on closed-loop LIBERO-Long success. Downstream architecture and training are held fixed within each interaction program.

Video initialization	VTA (%)	Joint (%)
Random initialization	20.0	27.2
Original Wan2.2	68.4	62.2
Robot-video pretrained	97.8	96.6

Table 13. Effect of the Mixture-of-Transformers architecture on closed-loop LIBERO-Long success. All variants use robot-video-pretrained initialization.

Architecture	VTA (%)	Joint (%)
Non-MoT	92.8	93.6
MoT	97.8	96.6

D. Ablations

D.1. Effect of Video-Backbone Initialization

We further examine whether the benefit of robot-video pre-training is specific to VTA or persists across interaction programs. Under otherwise matched downstream training, we initialize the video backbone either randomly, from the original Wan2.2 checkpoint, or from our robot-video-pretrained causal checkpoint. Table 12 shows a consistent ordering for both VTA and Joint. Relative to Wan2.2 initialization, robot-video pretraining improves success by 29.4 percentage points for VTA and 34.4 points for Joint, reaching 97.8% and 96.6%, respectively. Random initialization performs substantially worse for both programs. These results indicate that the benefit of the robot-video-pretrained backbone transfers across distinct video-action interaction structures.

D.2. Effect of the Mixture-of-Transformers Architecture

We isolate the effect of the Mixture-of-Transformers (MoT) parameterization while keeping the pretrained visual foundation fixed. Both variants start from the same robot-video-pretrained video checkpoint and use the same downstream data and training procedure. In the non-MoT variant, video and action tokens are processed by the same pretrained DiT; the MoT variant retains the pretrained video expert and introduces a separate action expert initialized from width-adapted copies of the corresponding video-expert layers. As shown in Table 13, the shared-DiT model already achieves strong performance, reaching 92.8% for VTA and 93.6% for Joint. MoT further improves success to 97.8% and 96.6%, corresponding to gains of 5.0 and 3.0 percentage points, respectively.

Table 14. Policy and dynamics accuracy on LIBERO-Long. LC-IDM and LC-FDM are independently trained specialist checkpoints; the multi-objective model trains policy, IDM, and FDM within a single checkpoint. Lower is better for MSE and IDM error; higher is better for success and SSIM.

Model	Policy	Real demonstrations				Counterfactual transitions			
	Succ. (%)	FDM MSE ↓	FDM SSIM ↑	IDM pos. (cm) ↓	IDM rot. (°) ↓	FDM MSE ↓	FDM SSIM ↑	IDM pos. (cm) ↓	IDM rot. (°) ↓
OPENWAM, LC-IDM, CF-only	–	–	–	1.76	2.39	–	–	1.17	3.60
OPENWAM, LC-IDM, mixed	–	–	–	0.44	1.21	–	–	1.22	3.68
OPENWAM, LC-FDM, CF-only	–	0.00734	0.9126	–	–	0.00940	0.8889	–	–
OPENWAM, LC-FDM, mixed	–	0.00186	0.9776	–	–	0.00962	0.8859	–	–
OPENWAM, multi-objective	92.8	0.00058	0.9934	0.43	1.10	0.01416	0.8398	1.73	3.55
UVA	88.0	0.00279	0.9645	2.03	1.60	0.01514	0.8134	3.57	12.99

E. External Dynamics Comparison

We compare OPENWAM with UVA [44], the closest released LIBERO system in our related work that exposes policy, inverse-dynamics, and forward-dynamics objectives within one model. Table 14 covers two settings: independently trained local-context dynamics components and a single multi-objective checkpoint trained jointly across policy, IDM, and FDM objectives.

For FDM, we report RGB prediction error in a common agent-view image space. For IDM, the two systems use different action parameterizations, so we compare the resulting end-effector trajectory error after executing the predicted actions from the same simulator state.

E.1. Independent Dynamics Components

The LC-IDM and LC-FDM specialists are trained either on counterfactual transitions only or on a 60/40 mixture of counterfactual and demonstration transitions. CF-only specialists are strongest on the held-out counterfactual distribution, while mixed supervision substantially improves fidelity on real demonstrations. Both remain well ahead of UVA on the counterfactual dynamics metrics.

E.2. Unified Policy and Dynamics Modeling

UVA trains policy generation, inverse dynamics, and forward dynamics within a single model. We therefore also evaluate one OPENWAM checkpoint trained with 60% real joint video-action supervision, 20% local-context IDM supervision, and 20% local-context FDM supervision; the IDM and FDM portions are each split equally between demonstration and counterfactual transitions.

The unified OPENWAM checkpoint outperforms UVA on policy success and on the evaluated IDM and FDM metrics. Its counterfactual dynamics accuracy is weaker than that of the dedicated specialists, showing the expected tradeoff between multi-objective training and specialization. The unified OPENWAM checkpoint also gives the lowest IDM and FDM errors on real demonstrations. These numbers measure fit to the demonstration distribution, not held-out generalization, because some of the same demonstrations may also be seen by the policy objective during training. Its policy score is also lower than the policy-specialized OPENWAM models as only 60% of training updates optimize the policy objective.